

Fundamentals of Information Systems Security

Lab 4: Network Traffic Analysis with Wireshark

Objective: Capture and analyze live HTTP and DNS traffic to understand protocol behavior.

Prerequisites

- Kali Linux VM.
- Wireshark installed (comes pre-installed in Kali).
- A web browser (e.g., Firefox).

Learning Objectives

By the end of this lab, you will be able to:

- Capture live network traffic using Wireshark.
- Apply display filters for HTTP, DNS, and TCP.
- Follow TCP streams to reconstruct HTTP requests.
- Analyze DNS queries and responses.

Lab Steps

Step 1: Launch Wireshark

Open the terminal and run:

```
sudo wireshark
```

Note: Root privileges are required to capture packets.

Step 2: Start Capture

Select the correct network interface (e.g., `eth0`, `wlan0`, or `any`). Click the blue shark fin to start capturing.

Step 3: Generate HTTP Traffic

Open a web browser and navigate to a plain HTTP website (e.g., `http://example.com` or `http://neverssl.com`). *Note: Avoid HTTPS sites for this lab as the payload is encrypted.*

Step 4: Generate DNS Traffic

In the terminal, run:

```
nslookup google.com  
nslookup microsoft.com
```

Step 5: Apply Filters

In the Wireshark filter bar, type `http` and press Enter. Only HTTP packets will be displayed.

To filter DNS traffic, type `dns`.

To filter for a specific domain: `dns.qry.name contains "google"`.

Step 6: Follow TCP Stream

Right-click on any HTTP packet and select **Follow** → **TCP Stream**. Observe the raw HTTP GET request and the server's HTTP response (including the HTML body).

Step 7: Analyze DNS Details

In the DNS packets, expand the **Domain Name System (response)** section in the packet details pane. Look at the **Answers** field to see the resolved IP addresses.

Step 8: Stop Capture and Save

Click the red square to stop the capture. Save the capture file as `lab_capture.pcapng` for later analysis.

Step 9: Replay/View Saved Capture

Open a saved capture file:

```
File -> Open -> lab_capture.pcapng
```

Optional: ARP and ICMP Analysis

- Filter for `arp` to see address resolution requests.
- Send a ping to a target (`ping 8.8.8.8`) and filter for `icmp` to see echo requests/replies.

Deliverables

Submit a report containing:

1. A screenshot of the Wireshark interface showing HTTP packets filtered.
2. A screenshot of the TCP stream dialog showing the HTTP GET request.
3. A screenshot showing the DNS query and response details (IP address resolved).
4. A brief explanation of the difference between HTTP and HTTPS traffic from a packet analysis perspective.